Jesse Cambon

Machine Learning Engineer | NLP & LLM Specialist

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SUMMARY

Machine Learning Engineer with deep experience in natural language processing (NLP), large language models (LLMs), and scalable cloud deployments. Creator of the popular open-source R package tidygeocoder (300K+ downloads). Proven track record of building impactful ML systems across healthcare and legal domains.

- **Skills**: Machine Learning • NLP • LLMs • Software Development • Data Science • Statistics
- **Tools**: Python • R • SQL • Git • Linux • Docker • Kubernetes • AWS • Azure

EXPERIENCE

Lead Machine Learning Engineer • Booz Allen Hamilton Aug 2023 – Present

- Key machine learning developer on a RAG chatbot used by the FDA for querying drug and medical device guidance documents. Led deployment of the app to AWS and implemented key features such as chat history.
- Built a generative AI application used by FDA supply chain analysts to summarize and analyze FDA premarket documents. Utilized AWS bedrock.
- Developed a framework for fine-tuning transformer models (PyTorch, CUDA, Kubernetes) to extract information of interest to FDA supply chain analysts from FDA premarket documents.

Senior Data Scientist • Lighthouse Mar 2022 – Aug 2023

- Built a framework for multi-GPU LLM training and inference for the classification of documents in legal cases with PyTorch, DeepSpeed, and Azure. The framework enabled rapid experimentation and the use of larger models.

Senior Data Scientist • Embedded Healthcare (acquired by Clarify Health) Mar 2021 – Feb 2022

- Spearheaded analytics to quantify the impact of clinical interventions on healthcare utilization metrics.
- Developed robust data pipelines and QA algorithms to support critical performance dashboards and analysis.

Lead Data Scientist • Booz Allen Hamilton Dec 2017 – Mar 2021

- Served as a technical lead and interim project lead. Managed 1-3 members of a data science team.
- Led the design and implementation of an NLP application that matched employees to internal job opportunities using word embeddings. The application was deployed for use by all employees and hiring managers.
- Overhauled and redesigned a python AWS Lambda application for enterprise-wide time charge accounting to provide efficient testing with AWS SAM, improved error handling, and smaller dependencies.
- Constructed regression models with Census microdata to analyze characteristics of metropolitan areas that attract immigrants. Presented at the 2018 Joint Statistical Meetings conference in Vancouver on behalf of DHS.

Analytics Consultant • Humana Aug 2014 – Nov 2017

- Built a predictive model to identify social isolation in a population of 3 million Medicare members.
- Performed observational and randomized studies to measure the impact of clinical interventions.
- Published research in the American Health & Drug Benefits Journal on out-of-network and ER utilization.
- Performed survey design and analysis that was featured in a Harvard School of Public Health case study.

PROJECTS

- Created the R package tidygeocoder which is widely used (300K+ downloads) in academia and industry for geocoding API calls. Published in the Journal of Open Source Software (2021).

EDUCATION

- **University of Louisville • M.Eng. & B.S**, Mechanical Engineering, highest honors
- University of Illinois / Coursera • Accelerated Computer Science Fundamentals Specialization